Machine Learning

Lecture 6: From Curvature to Logistic Regression

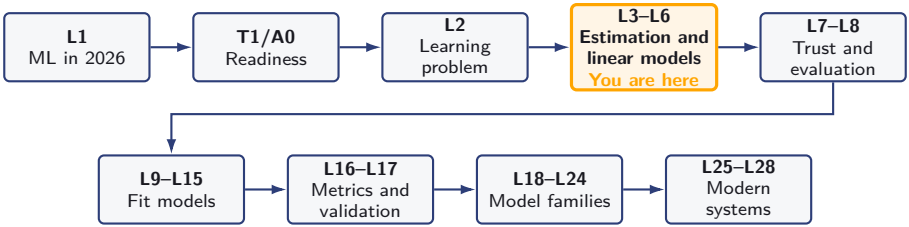
Instructor: Prof. Udit Bhatia

Civil Engineering; Computer Science and Engineering

Semester I, 2026



Course Roadmap: You Are Here

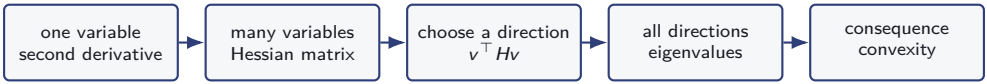


Lecture 5 connected the normal equation, eigendirections, and PCA. Today we first build the Hessian and curvature test from ordinary derivatives, then derive logistic regression from probability to loss.

Before Classification: One Mathematical Tool Is Missing

Linear regression gave us an objective $J(\theta)$ and a gradient. Before moving to logistic regression, we need to answer a new question:

How do we describe curvature when θ has many components?



Summary

No Hessian result is assumed. The next six slides build the complete argument from ordinary one-variable derivatives.

Learning Objectives

By the end of this lecture, you should be able to:

- connect directional curvature $v^\top Hv$ with eigenvalues of H;
- distinguish convexity, strict convexity, and uniqueness;
- derive $p \to$ odds $\to$ log-odds $\to$ sigmoid without a memorized jump;
- derive binary cross-entropy from Bernoulli likelihood;
- derive $\nabla J = X^\top(p - y)$ and $\nabla^2 J = X^\top WX$;
- prove logistic loss is convex and state when the minimizer is unique;
- separate probability estimation from the final decision threshold.

Notation: Observed, Chosen, And Computed Objects

Object	Shape	Meaning	Status
$X = [x_1^\top; \ldots; x_n^\top]$	$n \times d$	design matrix	observed
$y_i \in \{0, 1\}$	scalar	binary outcome	observed
θ	$d \times 1$	model parameters	learned
$z_i = x_i^\top \theta$	scalar	unrestricted linear score	computed
$p_i = \Pr(Y_i = 1 \mid x_i)$	scalar in $(0, 1)$	event probability	modeled
$p = (p_1, \ldots, p_n)^\top$	$n \times 1$	fitted probabilities	computed
$W = \mathrm{diag}(p_i(1 - p_i))$	$n \times n$	curvature weights	computed
$H = \nabla^2 J(\theta)$	$d \times d$	parameter-space curvature	computed

Summary

y_i is the observed label. z_i is an internal score. p_i is a modeled probability. They are three different objects.

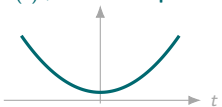
Step 1: Curvature In One Variable

For a scalar function $f(t)$, the first derivative is the slope and the second derivative tells us how that slope changes:

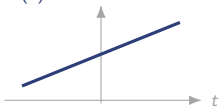
$$f'(t) = \frac{df}{dt}, \qquad f''(t) = \frac{d}{dt} f'(t).$$

For $f(t) = at^2 + bt + c$, we have $f''(t) = 2a$.

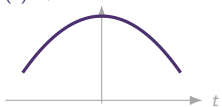
$f''(t) > 0$: bends upward



$f''(t) = 0$: no curvature



$f''(t) < 0$: bends downward



Summary

In one variable, $f''(t) \geq 0$ everywhere is the familiar curvature test for a twice-differentiable convex function.

Step 2: Build The Hessian Matrix

Let $\theta = (\theta_1, \ldots, \theta_d)^\top$. The gradient stores the d first partial derivatives, while the Hessian stores every second partial:

$$\nabla J(\theta) = \begin{bmatrix} \partial J/\partial \theta_1 \\ \vdots \\ \partial J/\partial \theta_d \end{bmatrix}, \qquad H(\theta) = \nabla^2 J(\theta) = \begin{bmatrix} \dfrac{\partial^2 J}{\partial \theta_1^2} & \cdots & \dfrac{\partial^2 J}{\partial \theta_1 \partial \theta_d} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial^2 J}{\partial \theta_d \partial \theta_1} & \cdots & \dfrac{\partial^2 J}{\partial \theta_d^2} \end{bmatrix}.$$

How to read the entries

Diagonal entries measure curvature along coordinate axes. Off-diagonal entries measure how changing one parameter alters the slope with respect to another.

Concrete two-parameter example

For

$$J(\theta_1, \theta_2) = 2\theta_1^2 + 2\theta_1\theta_2 + \theta_2^2,$$

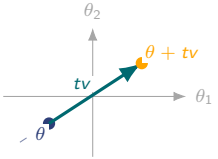
$$\nabla J = \begin{bmatrix} 4\theta_1 + 2\theta_2 \\ 2\theta_1 + 2\theta_2 \end{bmatrix}, \qquad H = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}.$$

Step 3: Reduce A Multivariable Move To One Variable

Choose a unit direction $v \in \mathbb{R}^d$ and trace a line through the current parameter vector:

$$\theta(t) = \theta + tv, \qquad g(t) = J(\theta(t)) = J(\theta + tv).$$

Differentiate the ordinary one-variable function $g(t)$ twice. Keep the vector v fixed.



$$g'(t) = \sum_{k=1}^{d} \frac{\partial J}{\partial \theta_k}(\theta + tv)\, \frac{d(\theta_k + tv_k)}{dt}$$

$$= \sum_{k=1}^{d} \frac{\partial J}{\partial \theta_k}(\theta + tv) v_k = \nabla J(\theta + tv)^\top v,$$

$$g''(t) = \sum_{j=1}^{d} \sum_{k=1}^{d} v_j \frac{\partial^2 J}{\partial \theta_j \partial \theta_k}(\theta + tv) v_k$$

$$= v^\top H(\theta + tv) v.$$

$\boxed{g''(0) = v^\top H(\theta) v}$ is the curvature seen while moving along v.

Step 4: Taylor's Quadratic Approximation

Apply ordinary one-variable Taylor expansion to $g(t) = J(\theta + tv)$ at $t = 0$:

$$g(t) \approx g(0) + tg'(0) + \frac{1}{2}t^2 g''(0).$$

Using $g'(0) = \nabla J(\theta)^\top v$, $g''(0) = v^\top H v$, and $\Delta = tv$ gives

$$\boxed{J(\theta + \Delta) \approx J(\theta) + \nabla J(\theta)^\top \Delta + \frac{1}{2}\Delta^\top H(\theta)\Delta.}$$

First-order information

$\nabla J(\theta)^\top \Delta$ predicts the initial rise or fall along Δ.

Second-order information

$\Delta^\top H \Delta$ measures how the slope changes along Δ. It is a scalar, not the vector $H\Delta$.

The Hessian appears after a direction turns the multivariable objective into an ordinary one-variable function.

Step 5: Why Can We Write $v = Qa$?

Because H is symmetric, its unit eigenvectors $q_1, \ldots, q_d$ form an orthonormal basis. Put these basis directions into the columns of

$$Q = [q_1 \ \cdots \ q_d], \qquad Q^\top Q = QQ^\top = I.$$

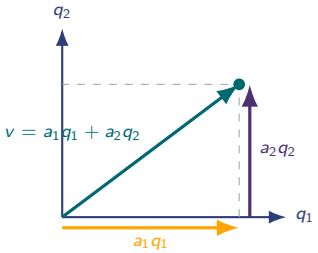
Any direction v can therefore be assembled from the q_j directions. Its signed amount along q_j is

$$a_j = q_j^\top v, \qquad a = Q^\top v.$$

Multiplying the basis matrix by these coordinates reconstructs the direction:

$$Qa = [q_1 \ \cdots \ q_d] \begin{bmatrix} a_1 \\ \vdots \\ a_d \end{bmatrix} = \sum_{j=1}^{d} a_j q_j = v.$$

v is the actual direction; a is the coordinate list for the same direction in the eigenvector axes.



Q supplies the axes.
a supplies the signed lengths.

Step 6: Eigenvalues Test Every Direction

When J has continuous second partial derivatives, H is symmetric. Therefore it has orthonormal eigenvectors q_j with $Hq_j = \lambda_j q_j$ and

$$H = Q\Lambda Q^\top.$$

Now substitute $v = Qa$ into $v^\top Hv$.
Use $Q^\top Q = I$ and $\|v\|_2 = 1$.

Because $Q^\top Q = I$ and $v^\top v = a^\top a = \sum_j a_j^2 = 1$,

$$\begin{aligned} v^\top Hv &= (Qa)^\top (Q\Lambda Q^\top)(Qa) \\ &= a^\top \Lambda a \\ &= \sum_{j=1}^{d} \lambda_j a_j^2. \end{aligned}$$

Thus directional curvature is a weighted average of the eigenvalues, with nonnegative weights a_j^2 that sum to one.

$$v^\top Hv \geq 0 \text{ for every } v \iff \lambda_j \geq 0 \text{ for every } j.$$

Convex Can Still Have Many Minimizers

The eigenvalue test separates two cases:

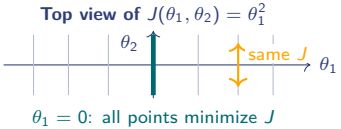
$$H \succeq 0 \iff \lambda_j \geq 0 \text{ for every } j.$$

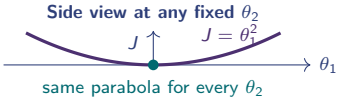
Convex

$\lambda_j \geq 0$: zero eigenvalues and flat directions are allowed.

Strictly convex

$\lambda_j > 0$: every direction has positive curvature, so there is at most one minimizer.





A zero eigenvalue creates a flat direction: the objective is convex, but an entire line can minimize it.

The Bridge: Reuse The Curvature Test, Do Not Restart

	Linear regression	**Logistic regression today**
outcome	continuous y	binary $y \in \{0, 1\}$
probability model	Gaussian	Bernoulli
mean model	$X\theta$	$p = \sigma(X\theta)$
loss	squared error	to be derived
gradient	$X^\top(X\theta - y)$	to be derived
Hessian	$X^\top X$	to be derived
curvature test	$v^\top H v \geq 0$	exactly the same test

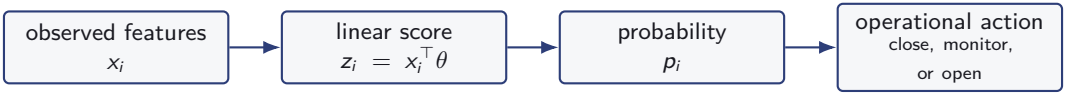
Question

What must change when the observation is binary, and which optimization ideas can remain unchanged?

Running Example: Will This Road Segment Become Impassable?

For each road segment and forecast window, observe

$$x_i = \begin{bmatrix} 1 & \text{rainfall} & \text{slope} & \text{drainage deficit} \end{bmatrix}^\top, \qquad y_i \in \{0, 1\}.$$



Warning

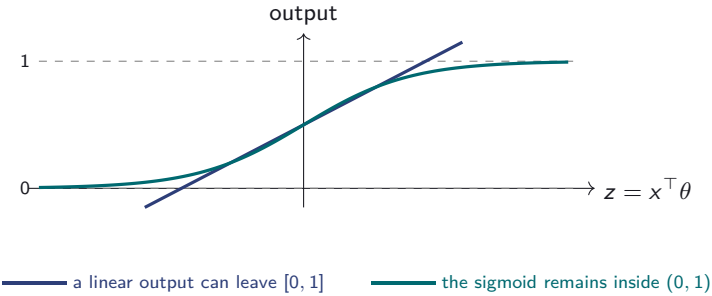
The model estimates a probability. The authority chooses an action using a threshold and the relative costs of missed closures and false alarms.

Why The Linear Score Cannot Itself Be A Probability

The familiar expression $z_i = x_i^\top \theta$ can take any real value:

$$z_i \in (-\infty, \infty).$$

A probability must satisfy $0 \leq p_i \leq 1$.



We need a principled map from an unrestricted score to a valid probability.

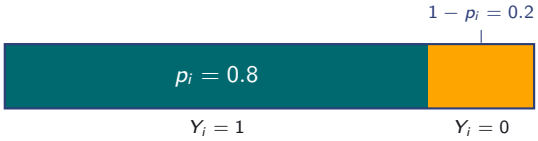
Start With Probability, Before Introducing Odds Or Sigmoid

For one observation, define

$$p_i = \Pr(Y_i = 1 \mid x_i).$$

Since the only outcomes are 0 and 1,

$$\Pr(Y_i = 0 \mid x_i) = 1 - p_i.$$



Summary

At this point there is no z_i, no sigmoid, and no loss. There are only the probabilities of the two possible outcomes.

Probability To Odds: Compare Event With Non-Event

For $p_i = 0.8$, compute the odds and interpret the result in ordinary words.

$$\text{odds}_i = \frac{\Pr(Y_i = 1 \mid x_i)}{\Pr(Y_i = 0 \mid x_i)} = \boxed{\frac{p_i}{1 - p_i}}.$$

For $p_i = 0.8$,

$$\text{odds}_i = \frac{0.8}{0.2} = 4.$$

Probability 0.8 means 80 events among 100 comparable cases. Odds 4 mean four events for every one non-event.

Summary

Probability compares events with all cases; odds compare events directly with non-events. They carry the same information on different scales.

Why Log-Odds? Match The Range Of A Linear Score

Odds are positive, but a linear score may be negative, zero, or positive. The logarithm creates matching ranges:

$$p \in (0,1) \xrightarrow{p/(1-p)} (0,\infty) \xrightarrow{\log} (-\infty,\infty).$$



$p = 0.5 \Rightarrow \log[p/(1-p)] = 0.$

Swapping event and non-event changes only the sign.

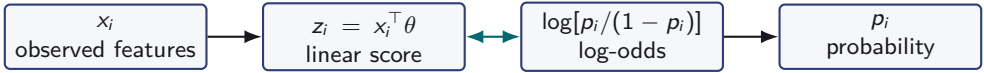
Only Now Introduce The Linear Score

The observed label remains $y_i \in \{0, 1\}$. Define the model's internal score

$$\boxed{z_i = x_i^\top \theta \in \mathbb{R}.}$$

Both z_i and the log-odds range over the full real line. Logistic regression makes the modeling choice

$$\boxed{\log \frac{p_i}{1 - p_i} = z_i = x_i^\top \theta.}$$



Warning

This is not the additive-noise model $Y = x^\top \theta + \varepsilon$ used for linear regression. We are modeling the conditional probability of a binary outcome.

Solve The Log-Odds Equation: The Sigmoid Appears

Starting from

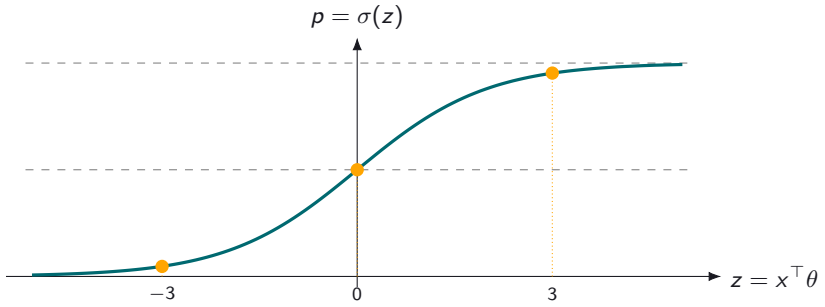
$$\log \frac{p_i}{1-p_i} = z_i,$$

solve for p_i without skipping algebra.

$$\begin{aligned}
\log \frac{p_i}{1-p_i} &= z_i \\
\frac{p_i}{1-p_i} &= e^{z_i} \\
p_i &= e^{z_i}(1-p_i) \\
p_i &= e^{z_i} - e^{z_i} p_i \\
p_i(1+e^{z_i}) &= e^{z_i} \\
p_i &= \frac{e^{z_i}}{1+e^{z_i}} = \frac{1}{1+e^{-z_i}} \\
&= \boxed{\sigma(z_i)}.
\end{aligned}$$

The sigmoid is the inverse of the logit, not an arbitrary decorative function.

Read The Sigmoid As A Probability Model



$z = -3 \Rightarrow p \approx 0.047$ $\quad$ $z = 0 \Rightarrow p = 0.5$ $\quad$ $z = 3 \Rightarrow p \approx 0.953$

$z = 0$ means equal odds and probability $1/2$. Adding one unit to z multiplies the odds by e.

Interactive explorer: move the score and compare probability, odds, and log-odds ↗

One Bernoulli Formula Covers Both Possible Labels

For $y_i \in \{0, 1\}$,

$$\boxed{\Pr(Y_i = y_i \mid x_i, \theta) = p_i^{y_i}(1 - p_i)^{1-y_i}.}$$

If $y_i = 1$,

$$p_i^1(1 - p_i)^0 = p_i.$$

The observation rewards a large p_i.

If $y_i = 0$,

$$p_i^0(1 - p_i)^1 = 1 - p_i.$$

The observation rewards a small p_i.

Question

If an event occurs but the model assigned $p_i = 0.01$, should that observation contribute a small or a large training penalty?

From Individual Probabilities To The Dataset Likelihood

Assume observations are conditionally independent given their feature vectors. Then

$$L(\theta) = \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1 - y_i}.$$

$y_1 = 1$ contribution p_1 $\times$ $y_2 = 0$ contribution $1 - p_2$ $\times$ $y_3 = 1$ contribution p_3 $\times \cdots \times$ y_n contribution selected by y_n

Warning

Conditional independence is across observations after conditioning on their inputs. It is not a claim that predictor columns are independent.

Take Logs: Product Becomes A Sum

Take the logarithm of the likelihood and show both logarithm rules being used.

$$\log L(\theta) = \log \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1-y_i}$$

$$= \sum_{i=1}^{n} \log \left[p_i^{y_i} (1 - p_i)^{1-y_i} \right]$$

$$= \sum_{i=1}^{n} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right].$$

We used

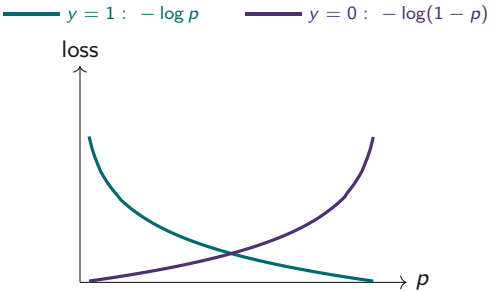
$$\log \prod_i a_i = \sum_i \log a_i, \qquad \log(a^b) = b \log a.$$

Maximum likelihood chooses θ that makes the observed labels collectively most probable.

Negate The Log-Likelihood: Binary Cross-Entropy

Optimization is usually written as minimization, so define

$$J(\theta) = -\log L(\theta) = -\sum_{i=1}^{n} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right].$$



Three names, one objective

Bernoulli negative log-likelihood, binary cross-entropy, and logistic loss refer to this same training objective.

Warning

Confidently wrong probabilities receive a very large penalty.

Substitute The Sigmoid: Compute The Log Terms First

With $p_i = e^{z_i}/(1 + e^{z_i})$, derive $1 - p_i$ before simplifying the loss.

Compute $1 - p_i$, $\log p_i$, and $\log(1 - p_i)$ one line at a time.

$$1 - p_i = 1 - \frac{e^{z_i}}{1 + e^{z_i}} = \frac{1}{1 + e^{z_i}},$$

$$\log p_i = \log \frac{e^{z_i}}{1 + e^{z_i}} = z_i - \log(1 + e^{z_i}),$$

$$\log(1 - p_i) = \log \frac{1}{1 + e^{z_i}} = -\log(1 + e^{z_i}).$$

One-Observation Loss: Simplify Without Skipping A Line

For one observation,

$$\ell_i = -y_i \log p_i - (1 - y_i) \log(1 - p_i).$$

Insert the two log expressions and simplify to a function of z_i and y_i.

$$\begin{aligned}
\ell_i &= -y_i[z_i - \log(1 + e^{z_i})] - (1 - y_i)[-\log(1 + e^{z_i})] \\
&= -y_i z_i + y_i \log(1 + e^{z_i}) + (1 - y_i) \log(1 + e^{z_i}) \\
&= -y_i z_i + [y_i + (1 - y_i)] \log(1 + e^{z_i}) \\
&= \boxed{\log(1 + e^{z_i}) - y_i z_i}, \qquad z_i = x_i^\top \theta.
\end{aligned}$$

Check Both Labels: Does The Compact Formula Behave Correctly?

$$\ell_i = \log(1 + e^{z_i}) - y_i z_i.$$

$y_i = 1$

$$\begin{aligned}\ell_i &= \log(1 + e^{z_i}) - z_i \\ &= \log(1 + e^{-z_i}).\end{aligned}$$

A large positive score gives a small loss.

$y_i = 0$

$$\ell_i = \log(1 + e^{z_i}).$$

A large negative score gives a small loss.

The log-odds model supplies p_i; the Bernoulli likelihood supplies the loss.

Interactive explorer: move one score and compare the two label losses ↗

Differentiate The Sigmoid Once, Then Reuse The Result

Differentiate $\sigma(z) = 1/(1 + e^{-z})$ and rewrite the derivative using only $\sigma(z)$.

$$\begin{aligned}\sigma'(z) &= -(1 + e^{-z})^{-2}(-e^{-z}) \\ &= \frac{e^{-z}}{(1 + e^{-z})^2} \\ &= \frac{1}{1 + e^{-z}} \frac{e^{-z}}{1 + e^{-z}} \\ &= \sigma(z)[1 - \sigma(z)].\end{aligned}$$

Therefore

$$\boxed{\frac{\mathrm{d}p_i}{\mathrm{d}z_i} = p_i(1 - p_i).}$$

The derivative is largest at $p_i = 1/2$ and tends to zero near 0 or 1.

Derive The Gradient: Score Residual Times Features

Use $\ell_i = \log(1 + e^{z_i}) - y_i z_i$ and $z_i = x_i^\top \theta$.

Differentiate first with respect to the scalar z_i, then apply the chain rule to the vector θ.

$$\begin{aligned}
\frac{\mathrm{d}\ell_i}{\mathrm{d}z_i} &= \frac{e^{z_i}}{1 + e^{z_i}} - y_i \\
&= p_i - y_i, \\
\nabla_\theta z_i &= x_i, \\
\nabla_\theta \ell_i &= \frac{\mathrm{d}\ell_i}{\mathrm{d}z_i} \nabla_\theta z_i \\
&= (p_i - y_i) x_i.
\end{aligned}$$

Summing observations,

$$\boxed{\nabla J(\theta) = \sum_i (p_i - y_i) x_i = X^\top (p - y).}$$

Derive The Hessian For One Observation

Differentiate $(p_i - y_i)x_i$ with respect to θ. Treat y_i and x_i as fixed observations.

$$\begin{aligned}\nabla_\theta^2 \ell_i &= \nabla_\theta[(p_i - y_i)x_i] \\ &= x_i(\nabla_\theta p_i)^\top, \\ \nabla_\theta p_i &= \frac{\mathrm{d}p_i}{\mathrm{d}z_i}\nabla_\theta z_i \\ &= p_i(1-p_i)x_i.\end{aligned}$$

Therefore

$$\boxed{\nabla_\theta^2 \ell_i = p_i(1-p_i)x_i x_i^\top.}$$

The scalar $p_i(1-p_i)$ determines how much curvature this observation adds.

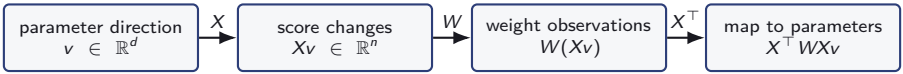
Stack The Hessian: Linear Algebra Returns

Define

$$W = \operatorname{diag}(w_1, \ldots, w_n), \qquad w_i = p_i(1 - p_i).$$

Adding the observation-wise Hessians gives

$$\boxed{\nabla^2 J(\theta) = \sum_{i=1}^{n} w_i x_i x_i^\top = X^\top W X.}$$



Summary

Unlike least squares, W changes with θ. Logistic curvature therefore changes as optimization moves.

Apply The Opening Theorem: Logistic Loss Is Convex

For any parameter direction v,

Expand $v^\top X^\top W X v$ as a sum and determine its sign.

$$
\begin{aligned}
v^\top \nabla^2 J(\theta) v &= v^\top X^\top W X v \\
&= (Xv)^\top W (Xv) \\
&= \sum_{i=1}^{n} p_i (1 - p_i)(x_i^\top v)^2 \\
&\geq 0.
\end{aligned}
$$

Every weight is nonnegative because $0 < p_i < 1$, and every square is nonnegative. Hence

$$\boxed{\nabla^2 J(\theta) \succeq 0 \quad \Longrightarrow \quad J \text{ is convex.}}$$

Convexity is not enough for a finite, unique estimate: rank deficiency creates flat directions, while complete separation can send coefficients toward infinity.

Why There Is Usually No Normal-Equation Formula

Linear regression sets

$$X^\top(X\theta - y) = 0 \quad \Longrightarrow \quad X^\top X\theta = X^\top y,$$

a linear equation in θ.

Logistic regression sets

$$\boxed{X^\top[p(\theta) - y] = 0,}$$

but $p(\theta) = \sigma(X\theta)$ is nonlinear in θ.

Gradient descent

$$\theta_{t+1} = \theta_t - \eta \nabla J(\theta_t).$$

Uses slope and a chosen step size.

Newton / IRLS step

$$\theta_{t+1} = \theta_t - H_t^{-1} \nabla J(\theta_t).$$

Uses local curvature $H_t = X^\top W_t X$.

Interactive explorer: compare first-order and curvature-scaled steps ↗

A Probability Is Not Yet A Decision

After fitting, an operational rule may use threshold τ:

$$\widehat{y}_i = \mathbb{I}[p_i \geq \tau].$$

Lower τ closes more roads:

- fewer missed impassable segments;
- more unnecessary closures.

Higher τ closes fewer roads:

- fewer unnecessary closures;
- more missed impassable segments.

Warning

Maximum likelihood estimates probabilities. The threshold expresses a decision cost and can be changed without refitting the probability model.

Interactive explorer: move the threshold and inspect the confusion matrix ↗

Linear And Logistic Regression: The Reusable Skeleton

	Linear regression	**Logistic regression**
response	continuous y	binary $y \in \{0, 1\}$
distribution	Gaussian	Bernoulli
mean model	$\mu = X\theta$	$p = \sigma(X\theta)$
loss	$\frac{1}{2}\|y - X\theta\|^2$	$-\sum[y \log p + (1-y)\log(1-p)]$
gradient	$X^\top(X\theta - y)$	$X^\top(p - y)$
Hessian	$X^\top X$	$X^\top WX$
closed form	sometimes	generally no
convexity test	$v^\top X^\top Xv \geq 0$	$v^\top X^\top WXv \geq 0$

Summary

Same workflow: choose a probability model, derive negative log-likelihood, differentiate, inspect curvature, optimize, then choose a deployment decision.

What You Should Be Able To Re-Derive

1. $v^\top H v = \sum_j \lambda_j a_j^2$ and the eigenvalue test for convexity;
2. $p \to p/(1-p) \to \log[p/(1-p)] = x^\top \theta \to p = \sigma(x^\top \theta)$;
3. Bernoulli likelihood $\to$ binary cross-entropy;
4. $\nabla J = X^\top (p - y)$ and $\nabla^2 J = X^\top W X$;
5. $v^\top X^\top W X v \geq 0$, hence convexity;
6. why convexity, uniqueness, finite coefficients, and a decision threshold are different questions.

Open the logistic-regression explorer ↗